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Ultrasound-Guided Percutaneous Cryoanalgesia for Pectus Excavatum: When Should It be Applied?

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Abstract

Introduction The addition of ultrasound-guided percutaneous cryoanalgesia (PCr) for pain management after pectus excavatum (PE) surgery offers a new and advantageous approach. Our aim is to describe our experience with PCr applied on the same day, 24 hours, and 48 hours prior to PE surgery.

Material and Methods Prospective pilot study in patients undergoing ultrasound-guided PCr (2019–2022) was divided into three groups: PCr on the same day of surgery (PCrSD), PCr 24 hours before (PCr24), and PCr 48 hours before (PCr48). We describe the application of technique and data obtained by comparing the three groups.

Results We present 42 patients (25 PCrSD, 11 PCr24, 6 PCr48). PCr24 had a shorter procedure duration than PCrSD (65.8 vs. 91.2 minute; $p = 0.048$). Related to analgesia, PCr24 and PCr48 showed lower opioid consumption than PCrSD in PCA volume (48.5 and 49.6 vs. 75.1 mL; $p = 0.015$) and PCA time (23.3 and 23.8 vs. 34.3 hours; $p = 0.01$). Degree of pain (VAS scale) on the day of surgery and on the second postoperative day was lower in PCr24 and PCr48 than in PCrSD (4 and 2 vs. 5; $p = 0.012$; 0 and 1 vs. 2; $p = 0.01$, respectively) as well as shorter hospital stay (3 and 3.5 vs. 5 days; $p = 0.021$). In addition, PCr24 showed lower opioid consumption and hospital stay than PCr48 ($p > 0.05$). The greatest savings in hospital costs were obtained in the PCr24 group.

Conclusion PCr48 and PCr24 prior to PE surgery offers lower opioid consumption, less pain and shorter hospital stay than PCrSD. PCr24 is comparable to PCr48, but seems to show advantages and simpler logistics for the patient and the hospital.

Keywords
- pectus excavatum
- percutaneous
- cryoanalgesia
- patient-controlled analgesia

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Introduction

Pectus excavatum (PE) is a thoracic defect relatively frequent and, although in most cases it is only an aesthetic defect, it can be the cause of serious cardiorespiratory problems. If indicated, after evaluation of physical, physiological, psychosocial, and anthropometric factors, surgical correction is recommended.[1,2]

The classic Ravitch technique was improved and largely replaced by Nuss' minimally invasive repair of pectus excavatum (MIRPE) in the 1990s.[3–5] This technique offers smaller

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incisions, avoids cartilage resection and osteotomy and is accompanied by shorter length of intervention.[6,7] However, the advantages of the new surgical technique were not accompanied by improvements in the intense and long-term postoperative pain that characterize this procedure and which results in high opioid consumption and long hospital stays.

Thoracoscopic cryoanalgesia has emerged as one of the most advantageous strategies for pain control in recent years. Numerous publications have demonstrated a lower degree of pain, lower opioid consumption, and shorter hospital stay.[8–12] The novel and recent ultrasound-guided percutaneous approach is an efficient, safe, and reproducible procedure. It has advantages over the thoracoscopic approach by avoiding possible lesions due to thermal contact of the cryoprobe, less dissection, less thoracoscopy time, and in case of having to repeat the technique due to failure of the first application, the approach is less complicated. Moreover, as the effect of cryoanalgesia can start between 6 and 48 hours after application, this approach allows it to be applied before surgery.[13]

With the aim of improving postoperative pain management in PE surgery, we decided to apply and describe our experience with ultrasound-guided percutaneous cryoanalgesia (PCr) applied on the same day, 24 hours, and 48 hours prior to PE surgery. As a secondary objective, we analyzed the economic impact on the hospital cost of the application of this novel technique.

Material and Methods

Study Design

A prospective study was performed including all patients undergoing Nuss repair and ultrasound-guided PCr from January 2019 to January 2022 at our institution. They were classified into three groups according to the time of PCr application: the same day of surgery (PCrSD), 24 hours (PCr24) or 48 hours before surgery (PCr48). All patients with age over 18 or less than 14 years, chest wall deformities other than PE (pectus carinatum, pectus arcuatus...), previous chest surgery, chronic use of analgesics or any anesthetic contraindication were excluded. Patients in whom more than one bar was placed were also excluded. The study was approved by the Institutional Review Board of our Hospital and conformed to the guidelines of the 1975 Declaration of Helsinki. Informed consent was signed by the parents or legal guardians of all patients at the outpatient clinic after the decision to be included in the surgical waiting list.

We described and analyzed demographic and clinical-radiological variables, operation time and PCr time, consumption of opioid analgesia in PCA (patient-controlled analgesia), degree of pain assessed by visual analog scale (VAS), length of hospital stay (LOS), intraoperative and postoperative complications, and need for oral analgesia 30 days after surgery.

Finally, the Accounting Department provided us with data on the economic analysis of the cost of procedures, material and hospital stay. With these data we analyzed the economic impact of PCr versus other strategies previously used in our patients (thoracic epidural catheter [EPI] and PCA [PCA]).

Percutaneous Cryoanalgesia Technique and Nuss Procedure

The procedures were performed by the same surgeon and anesthesiologist in all patients after verification of fasting, aptitude in the preanesthetic evaluation, and correct completion of the informed consent form. Patients in the PCr24 and PCr48 groups were admitted 24 or 48 hours before surgery to the Day Hospital, where they underwent PCr under sedation. Patients in the PCrSD group were admitted on the same day of surgery and PCr was performed in the operating room under general anesthesia.

Percutaneous cryoanalgesia: Before starting the procedure, the calibration of the machines and the good condition of all the material to be used were checked. The position chosen was lateral decubitus with 45 degree inclination. Five intercostal blocks were performed at the level of posterior costal angle in each hemithorax to completely cover the T3-T7 dermatomes. Under sterile technique and ultrasound control, an 18-gauge angiocatheter was introduced at a 30 degree angle until the desired intercostal space was reached. Once the 18-gauge needle was inserted, the needle was removed, leaving the canula, which serves as an introducer to facilitate insertion of the 21-gauge cryotherapy needle. Subsequently, the cryoanalgesia needle (needle model A-08/20/N/RF with rounded tip, 9 cm length, 7 mm freezing tip, and thickness of 21 with Metrum Cryoflex Cryo-S Electric II Painless equipment [Warsaw, Poland]) was introduced up to the intercostal nerve and a simple freezing cycle was applied for 2 minutes with carbon dioxide at −70 degrees, followed by 30 seconds of thawing. Finally, the needle and angiocatheter were remove and this same procedure was repeated from T3 to T7 in both hemithorax (►Fig. 1).

Nuss procedure: The PE was corrected with placement of a Nuss bar by thoracoscopy according to our standard technique. Sternal elevation with crane technique, two small incisions on each side of the chest wall, and a retrosternal space were created below the point of greatest sternal collapse; an introducer was advanced medially and exteriorized through the contralateral hemithorax. A Blake was placed as a guide for the placement of the bar. The bar was

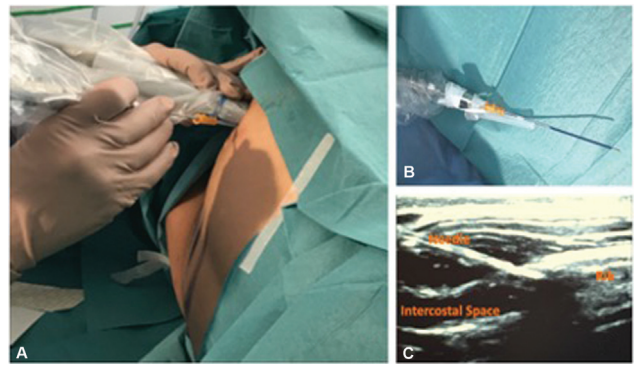

Fig. 1 Ultrasound-guided percutaneous cryoanalgesia procedure. (**A**, **B**) Cryoanalgesia needle through angiocatheter. (**C**) Ultrasound image of the intercostal space where the tip of the cryoanalgesia needle is placed.

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introduced with the convexity backward and then flipped so that the convexity was anterior, which elevated the sternum and corrected the PE. The bar was fixed on the left hemithorax with a stabilizer (MedXpert Universal System) and on the right side with pericostal stitches.

Postoperative Care

Postoperative care and analgesic protocol were established by the Pain Unit of our Hospital and were applied in the same way in all patients. After surgery, the patient was transferred to Anesthesia Resuscitation Unit for postoperative control and a PCA of fentanyl (0.5 µg/kg/mL, with a maximum of 25 µg/mL) and metamizole magnesium 4 mg/kg/mL, with a maximum of 200 mg/mL) was prescribed, with basal perfusion at 2 mL/h during the first 8 hours after surgery, at which time, depending on the analgesia achieved, it was lowered to 0.5 mL/h or left alone at 1 mL boluses. Subsequently, analgesic de-escalation was initiated, first with intravenous analgesia, then alternating intravenous and oral analgesia and finally maintaining only oral analgesia according to the degree of pain assessed with the VAS. Pain assessment was performed twice a day, once by the Pain Unit and once by the nursing staff. When pain was controlled with oral analgesia, the patient could walk independently and had good tolerance to enteral feeding; he was discharged home with acetaminophen, metamizole, and oral dexketoprofen on demand.

Outpatient visits were scheduled in the first and third postoperative weeks. Thereafter, outpatient visits were made every 2 months until 6 months postoperative and then every 6 months until bar removal. Each visit consisted of an anamnesis on the evolution, pain degree, possible complications, degree of satisfaction and analgesic intake. It was accompanied by a physical examination evaluating the incision sites, chest wall mobility, and aesthetic aspect.

Statistical Analysis

Data were collected in Microsoft Excel software version 16.61 (Redmond, Washington, United States) and analyzed with SPSS Statistic version 25 (Chicago, Illinois, United States). Kolmogorov–Smirnoff and Shapiro–Wilk tests were used to compare the distributions of numerical variables. Continuous variables normally distributed were expressed as means ± standard deviation, and those that were not normally distributed, as medians with interquartile range (Q1–Q3). Categorical variables were expressed as frequency (n) and percentage (%). Quantitative variables were compared with the analysis of variance (ANOVA) when they were normal and with the Kruskal–Wallis test when they were not normal. Qualitative variables were analyzed with the chi-square test corrected by Pearson. Statistical significance was established with a value of p-value of 0.05.

Results

A total of 42 patients were included (84% males, mean age 14.9 ± 0.9 years). There were 25 patients in PCrSD group, six in PCr24 and 11 in PCr48. The main reason for consultation was aesthetic (88.1%), followed by dizziness (23.8%), pain (7.1%), and dyspnea (4.8%) in all of them. Demographic, clinical, and radiological data are shown in ►Table 1.

Time of Surgical Procedure and PCr Application

The time of the surgical procedure and PCr application in the total number of cases, was 66.1 ± 15.8 minutes and 83.9 ± 28.6 minutes, respectively. No statistically significant differences were found in surgical procedure time between the groups (66.5 ± 17.5 minutes in PCrSD, 65.3 ± 18.8 minutes in PCr24, and 66.3 ± 8.2 minutes in PCr48; $p = 0.517$). However, we did find a shorter PCr application time in the PCr24 group compared with the PCrSD group, and this difference was statistically significant (65.8 ± 7.9 vs. 91.2 ± 33.8 minutes; $p = 0.048$, respectively).

Opioid Use and Degree of Pain (VAS)

The consumption of opioid analgesia in hours and volume, was significantly lower in the PCr24 and PCr48 groups compared with the PCrSD group (23.3 ± 1.6 and 23.8 ± 1.8 vs. 34.3 ± 15.3 hours; $p = 0.01$; 48.5 ± 6.5 and 49.6 ± 6.6 vs. 75.1 ± 34.2 mL; $p = 0.015$, respectively). There were no differences in the number of boluses applied between the groups. Similarly, the degree of pain on the day of surgery and on the second postoperative day was significantly lower in the PCr24 and PCr48 groups than in the PCrSD group (4 and 2 vs. 5; $p = 0.012$; 0 and 1 vs. 2; $p = 0.01$, respectively). The data collected on the consumption of analgesics and degree of pain measured with the analog scale (VAS) in the first five postoperative days are shown in ►Table 2.

Length of Hospital Stay (LOS)

The median hospital stay was shorter in the PCr24 and PCr48 groups versus PCrSD, and these findings were statistically significant (3 and 3.5 vs. 5 days; $p = 0.021$, respectively) (►Fig. 2).

Complications and Follow-up

No serious complications were reported during the surgical procedure, PCr or during follow-up. As minor complications, two patients in the PCr48 group and one in the PCrSD group developed self-limited hematomas after application of cryoanalgesia. In total, 95.8% presented nausea and vomiting related to opioid medication (68% in PCrSD, 33.3% in PCr24, and 36.4 in PCr48); 11.9% required bladder catheterization after developing acute urinary retention (two in PCrSD and three in PCr24); one patient in the PCrSD group developed pleural effusion that did not require chest tube placement for his resolution; and another patient in the same group had severe respiratory depression that was resolved with oxygen therapy with nasal glasses within 24 hours. No pneumothorax, bar displacement, or problems with surgical wounds were observed.

In the outpatient visits, all patients were satisfied with their esthetic image and none reported dyspnea, dizziness, or fatigue. Regarding pain control at home, the need for analgesia (paracetamol, ibuprofen, or metamizole oral) during the first 30 days was 90.5% (92% in PCrSD, 100% in PCr24 and

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Table 1 Demographic, clinical, and radiological characteristics

	PCrSD ($n = 25$)	PCr24 ($n = 6$)	PCr48 ($n = 11$)	*p*-Value
Age (years); average ± SD	14.9 ± 0.9	14.7 ± 1.3	15.2 ± 0.9	*0.505*
Body mass index, average ± SD	18.8 ± 1.7	20.7 ± 2.3	19.5 ± 2.8	*0.347*
Gender; *n* (%)				
Males	21 (84)	6 (100)	11 (100)	*0.222*
Females	4 (16)	0	0	
ASA; *n* (%)				
ASA I	9 (36)	3 (50)	2 (18.2)	*0.417*
ASA II	13 (52)	3 (50)	9 (81.8)	
ASA III	3 (12)	0	0	
Spirometry; *n* (%)				
Normal	24 (96)	6 (100)	10 (90.9)	*0.675*
Slight alteration	1 (4)	0	1 (9.1)	
Echocardiography; *n* (%)				
Normal	22 (88)	5 (83.3)	11 (100)	*0.492*
Slight alteration	3 (12)	1 (16.7)	0	
RM (dimensionless); average ± SD				
Haller index				
Inspiration	4.4 ± 1.8	4.5 ± 2	4.4 ± 1.4	*0.893*
Expiration	5.9 ± 2.9	6.8 ± 2	6.4 ± 3.3	*0.762*
Correction index				
Inspiration	34.2 ± 11.2	37.4 ± 11.6	30.9 ± 13.5	*0.540*
Expiration	41.8 ± 13.8	49.4 ± 7.7	37.5 ± 18.8	*0.291*

Abbreviations: ASA, American Society of Anestesthesiologists Physical Status; RM, magnetic resonance imaging; SD, standard deviation.

Table 2 Analgesia and degree of pain assessed by visual analog scale (VAS) during postoperative days (DPO)

	PCrSD ($n = 25$)	PCr24 ($n = 6$)	PCr48 ($n = 11$)	*p*-Value
Opioid consumption				
Hours PCA; average ± SD	34.3 ± 15.3	23.3 ± 1.6	23.8 ± 1.8	*0.01*
Total volume PCA (mL); average ± SD	75.1 ± 34.2	48.5 ± 6.5	49.6 ± 6.6	*0.015*
Bolus PCA (units); median (Q1–Q3)	15 (7–18)	7 (0.7–19.2)	11 (8–14)	*0.687*
Degree of pain (VAS)				
DPO 0; median (Q1–Q3)	5 (3–6.5)	4 (3–5.7)	2 (2–3)	*0.012*
DPO 1; median (Q1–Q3)	2 (1.5–5.5)	0.5 (0–1.7)	5 (1–6)	*0.055*
DPO 2; median (Q1–Q3)	2 (1–6)	0	1 (0–1)	*0.01*
DPO 3; median (Q1–Q3)	1 (0.25–2)		1 (0.7–1)	*0.043*

Abbreviations: IV, intravenous; O, oral; PCA, patient-controlled analgesia.

PCr48) in the first postoperative week, while in the third postoperative week only 38.7% (40% in PCrSD, 66.7% in PCr24, and 18.2% in PCr48) of the patients were taking analgesia on demand.

Costs and Savings

When comparing the data of patients who underwent PCr versus those who received epidural catheter (EPI) or PCA in our historical cohort, we found that PCr entailed an additional material cost of €629.2 per patient; and an additional cost per procedure of €1,319 per patient in the PCrSD group and €832.2 per patient in the PCr24 and PCr48 groups. However, the application of PCr resulted in savings in terms of reduced hospital stay, being more marked in the PCr 24 group. PCr24 resulted in savings of €6,218/patient compared with PCA, €5,495/patient compared with EPI, €2,602/patient compared with PCrSD, and €723/patient compared with PCr48.

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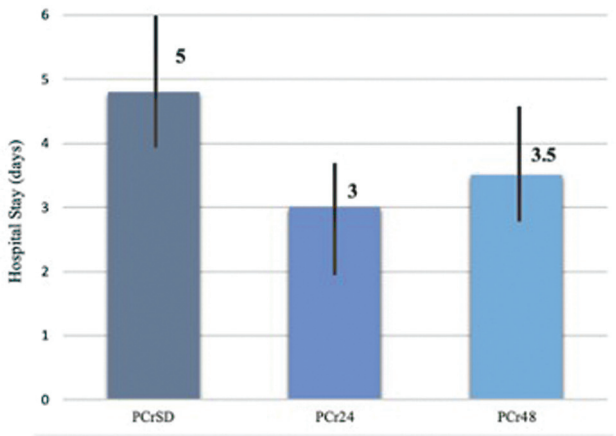

Fig. 2 Difference in LOS between the three groups. The number represents the median and the line marks the Q1 to Q3 of each group. LOS, length of stay.

Discussion

The main advantage of MIRPE over the Ravitch technique, is its lower invasiveness. Following this idea, ultrasound-guided PCr is offered as an alternative approach to its thoracoscopic application, contributing significantly to the minimally invasive character currently being pursued. In addition, PCr offers precise anatomical definition, high spatial resolution, good tissue contrast, real-time visualization, low cost and minimally invasive monitoring. It is a safe, reproducible technique with minimal risk of complications.[14,15] All this translates into numerous advantages over the thoracoscopic approach: it is more physiological for the patient, does not require selective intubation or one-lung ventilation, avoids the need for bilateral thoracoscopy and additional incisions, and avoids potential thermal injuries due to contact with the cryoprobe.[16] On the other hand, anesthesiologists are familiar with ultrasound-guided percutaneous techniques for infiltration of local anesthetics, making it an attractive technique with a short learning curve and great collaboration between the two groups.

Cold application to the intercostal nerves initiates a cascade of pathophysiological events that interrupt nerve conduction of painful stimulus without irreversible nerve damage while maintaining pain control for several weeks.[8] Several studies based on the application of thoracoscopic cryoanalgesia in MIRPE conclude a shorter hospital stay, reduced opioid consumption, and less pain compared with conventional techniques.[3,9–12,17–20] As already described in the literature, the effect of cryoanalgesia can vary between 6 and 48 hours after application. In 2021, we conducted a study comparing patients in whom PCr was applied on the same day of PE surgery and those in whom it was applied 48 hours before surgery. We obtained a similar efficacy and safety profile between the two strategies but with a lower degree of pain, lower opioid consumption, and shorter hospital stay by the PCr48 group.[13] For reasons mainly related to the COVID-19 pandemic, we started to apply it 24 hours before surgery. When comparing the three groups in this study, the PCr24 and PCr48 groups had a lower pain score, lower opioid consumption, and shorter hospital stay than the PCrSD group. Also, the data obtained from the PCr24 group related to these variables, offered advantages over those obtained in the PCr48 group, although the results did not reach statistical significance. We believe that the possible explanation lies in the small number of patients in the PCr24 group. In future studies, we hope to be able to compare larger groups of patients that would yield more solid results on the potential benefits of PCr24 versus PCr48.

The main disadvantage of cryoneurolysis is the duration of the procedure. In our series the mean PCr time was 83.9 ± 28.6 minutes. The PCr24 group showed a shorter application time compared with the PCrSD group, and this was statistically significant (65.8 ± 7.9 vs. 91.2 ± 33.8 minutes, respectively; $p = 0.048$). We believe that the reason for this was that the patients in whom PCr was applied 24 hours earlier were the last ones performed in the series and therefore the experience was much greater than with the first patients in whom it was applied on the same day of surgery. However, this time cost must be weighed against the benefits of a significantly prolonged analgesia, less pain, and opioid use and shorter hospital stay, which in global terms means a lower hospital cost.[21] Aiken et al analyzed the economic impact of thoracoscopic cryoanalgesia versus patients treated with epidural catheter (EPI) and PCA, concluding that the thoracoscopic cryoanalgesia group consumed fewer resources and therefore resulted in savings hospital costs.[22] Similarly, we compared our PCr groups with a historical cohort of patients treated with EPI and PCA, finding a saving justified by a shorter hospital stay in the PCr groups compared with the other two, with the savings being more pronounced in the PCr24. By repeating the process, more experience will be gained and the time to perform PCr will be shortened, further contributing to cost reduction.

The complications described are minimal (local hematomas), there is no evidence of permanent nerve damage or neuroma, and only one serious case of myonecrosis and abscess has been described.[15,23–28] In our series, there were no serious complications related to the technique. Nausea and vomiting were present in 95.8% of patients and most related it to opioid use. Eliminating the routine use of PCA on the first postoperative day would reduce these symptoms and may further shorten the hospital stay as described by Sujka et al, with a postoperative admission duration of 1.5 days in their series.[29] Our idea is to go a step further and eliminate PCA from the postoperative protocol in patients with CPr, to verify the decrease in adverse effects of opioids, reduction of hospital stay and, therefore, hospital costs. This idea is in line with the protocols established by Rettig et al or Di Fiore et al[30,31]

Finally, percutaneous ultrasound-guided cryoanalgesia (PCr) contributes to the acquisition of skills for its potential use in other pediatric surgical interventions requiring short- and long-term pain control (painful burns of long evolution, rib fractures, palliative process, and lung transplantation...).[14,15,32–40]

The main limitation of our study was the small sample size, especially in the PCr24 group. On the other hand, we do not know the possible effects and complications that may

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appear in a longer follow-up. Future revisions should include the evolution and follow-up of possible complications until the bar is removed.

Conclusion

We propose PCr for PE correction as a standardized model of postoperative analgesia because it is effective and long-lasting. The application of PCr24 or PCr48 prior to PE surgery offers lower opioid consumption, less pain, and shorter hospital stay than that applied on the day of surgery. PCr24 seems to show advantages and presents simpler logistics for the patient and the hospital center.

Conflict of Interest

None declared.

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